HOSPITAL CARDIOVASCULAR AI IMAGING CENTER

Epicardial & Pericoronary Fat AI Quantification Report

Automated 3D WholeBody SegResNet & CRISP-CT Coronary Inflammation Analysis

Patient ID: 0506
Exam Date: 2026-10-29
Modality: Cardiac CT (CCTA)

ELEVATED RISK

CRISP-CT Vascular Inflammation & Fat Burden: Elevated (~81th %ile)

Threshold: PVAT > -70.1 HU indicates active inflammation.

EPICARDIAL ADIPOSE TISSUE (EAT)

Total EAT Volume: 140.7 cm³

Mean Attenuation: -77.0 HU

Chamber Sub-Compartments:

- Left Ventricle (LV): 51.0 cm³
- Right Ventricle (RV): 63.7 cm³
- Left Atrium (LA): 22.5 cm³
- Right Atrium (RA): 28.7 cm³

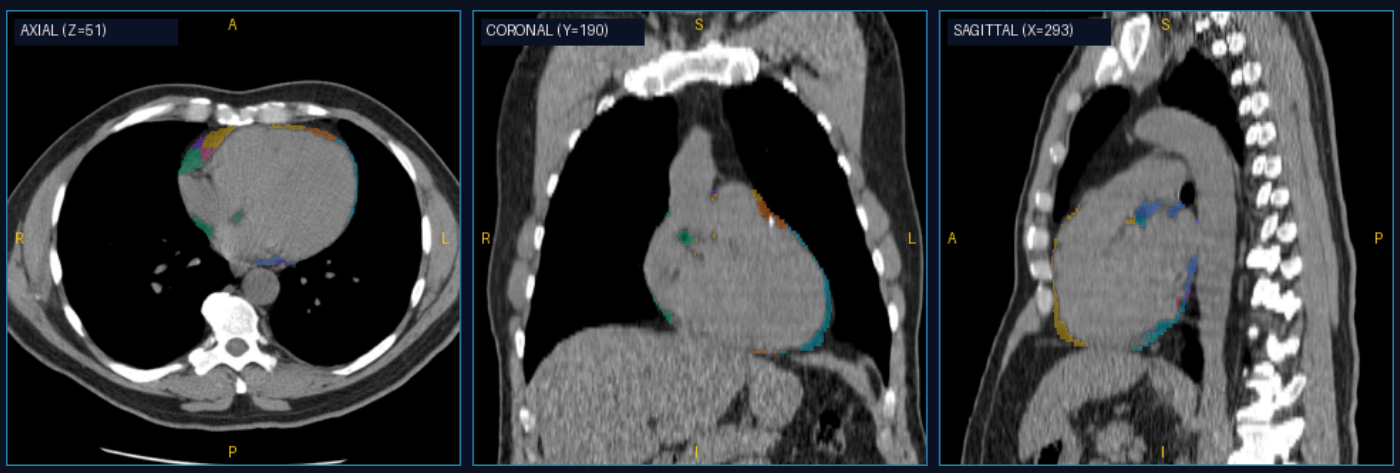
PERICORONARY ADIPOSE TISSUE (PVAT)

LAD PVAT:	13.1 cm³	-78.9 HU	Normal
LCX PVAT:	3.1 cm³	-62.0 HU	INFLAMED
RCA PVAT:	8.3 cm³	-72.2 HU	Normal

Total Pericoronary Fat Volume: 24.6 cm³

3-VIEW ORTHOGONAL MULTI-PLANAR RECONSTRUCTION (MPR) WITH AI OVERLAYS

Axial (Z), Coronal (Y), and Sagittal (X) cardiac planes with 8-compartment adipose masks



STRUCTURED CLINICAL IMPRESSION (READY FOR EHR / POWERSCRIBE):

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QUANTITATIVE CARDIAC ADIPOSE & INFLAMMATION AI ANALYSIS:
- Overall Biomarker Triage: ELEVATED RISK
- Total Epicardial Fat (EAT) Volume: 140.7 cm3 (Elevated burden, ~81th %ile)
- Total EAT Mean Attenuation: -77.0 HU
- Pericoronary Fat Attenuation (CRISP-CT active inflammation threshold: > -70.1 HU):
    * LAD PVAT: -78.9 HU - Normal Attenuation (· -70.1 HU)
    * LCX PVAT: -62.0 HU - ELEVATED INFLAMMATION (> -70.1 HU)
    * RCA PVAT: -72.2 HU - Normal Attenuation (· -70.1 HU)
IMPRESSION: Total Epicardial Adipose Tissue (EAT) volume is 140.7 cm³ (Elevated burden, ~81th percentile), with mean attenuat
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For Investigational / Research Use Only (RUO). Clinical verification required by board-certified physician.

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AI GENERATIVE NARRATIVE FINDINGS & CLINICAL IMPRESSION:

TECHNIQUE: Non-contrast CT.
FINDINGS:
Medical devices: None present.
CHEST:
Lungs and airways: No evidence of mass or focal consolidation. Central airways are patent.
Pleura: No pleural effusion or pneumothorax.
Mediastinum and hila: No evidence of size significant lymphadenopathy within the mediastinum.
Heart and pericardium: Normal heart size. Mild calcification of the coronary arteries. No pericardial effusion.
Great vessels: Major mediastinal vasculature is unremarkable.
Esophagus: No esophageal abnormality.
Chest wall and breast: Normal.
Bones: Moderate degenerative changes of the mid-thoracic spine. No suspicious bony abnormality identified.
IMPRESSION:
1. Mild coronary artery calcification.
2. No evidence of active coronary inflammation.

KEY FINDINGS: Coronary artery wall calcification

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